

UNIT – 4
ASSESSMENT – 1
SOLUTIONS

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TASK 1:

1. Inductive Learning

Inductive learning learns a general pattern from previously observed patient examples and uses it to predict diabetes risk for new patients.

Target Variable

Diabetes

0 → Non-Diabetic

1 → Diabetic

Important Features

- **Glucose** – important indicator of diabetes risk.
- **BMI** – higher BMI is associated with increased diabetes risk.
- **Age** – diabetes risk generally increases with age.
- **Blood Pressure**
- **Cholesterol**
- **Family History**

2. Handling Class Imbalance

The dataset may contain more non-diabetic than diabetic patients.

For this implementation, **class weighting** is used.

```
class_weight = "balanced"
```

This gives more importance to the minority diabetic class without artificially generating new patient records.

3. Train-Test Split

70% → Training

30% → Testing

A **stratified split** is used so that both classes maintain approximately the same proportion in training and testing data.

4. Preprocessing

- Numerical features → Standardization
- Family History → Binary encoding
- Missing values → Should be handled before training
- Target → Binary classification

TASK:2

The Decision Tree recursively divides patients according to the feature that provides the best separation between diabetic and non-diabetic cases.

For medical screening, **Recall is particularly important** because False Negatives represent diabetic patients who were incorrectly classified as non-diabetic.

Class weighting is used to give greater importance to diabetic cases.

TASK 3:

Logistic Regression is used because diabetes prediction is a binary classification problem.

The model estimates the probability that a patient belongs to the diabetic class.

The absolute values of Logistic Regression coefficients are used to determine feature importance.

TASK 4:

MDP

States

Low_Risk

Moderate_Risk

High_Risk

Critical_Risk

Actions

Diet

Exercise

Medication

Monitor

Reward

An action that improves the patient's health state receives a higher reward.

Health improves → Positive reward

Health unchanged → Small reward

Health worsens → Negative reward

Transition

Example:

High_Risk

|

Medication

↓

Moderate_Risk

The Q-Learning agent learns the action with the highest expected long-term reward.

